Planing:

https://miro.com/app/board/uXjVH04JbwU=/?share_link_id=802185247908
Using Miro, I planned out requirements -> analysis -> features -> development
(The link allows for commenting, feel free to comment on my Miro)

Tech Stack:

Frontend (React + Vite) - I have experience with React and Vite back when I was still exploring the world of IT/software, so naturally I chose to use them, although currently I am not a frontend guy.

Backend (Spring Boot) - beside Python, Java is the next programming language that I am familiar with since it has been some time since I used Python in academics, but a few revisions will recover my Python skills. Furthermore, I would also like to use Java to make a real application. On a technical level, Java is faster than Python, although in this challenge runtime shouldn't be an issue or at least a factor to consider.

Storage (Json) - Using Json, I simplified the database without using more advanced choices like MySQL, although I do acknowledge that using Json would mean bad data security. On a technical level, I would like the data to be "saved" such that if we were to run the application again, it would load previously saved data.

Architecture

User
↓
Frontend (JavaScript)
↓
HTTP/REST API Backend (Java Spring Boot)
↓
JSON File Storage

AI Usage:

AI is mostly used in helping give code ideas (such as how to call API etc), debugging. Frontends are all generated by AI.
By telling Ai to code the entire application out, and I ran a simple user test, after seeing that it works, then the real work begins.
Since I have the whole scaffold down, I read line by line and type it in on another workplace (the actual workplace to be submitted). By this way, I was able to go through the backend code without the risk of just "looking through" everything and vibe code the entire application.
(Refer to Ai_usage.md)

Flow Chart: